Virtual Shadow: Making Cross Traffic Dynamics Visible through Augmented Reality Head Up Display

Hyungil Kim
Department of Industrial and Systems Engineering, Virginia Tech.

Objective: This work aims to (1) design a novel driver interface for cross traffic alerts taking advantage of an augmented reality (AR) head-up display (HUD), (2) prototype design ideas for a specific use-case of pedestrian collision warning, and (3) evaluate usability in consideration of unique aspects of interaction with AR while driving.

Background: Current pedestrian collision warning systems use either auditory alarms or visual symbols to inform drivers. However, they cannot tell the driver where the detected pedestrians are located, which is critical for the driver to respond appropriately.

Method: Ecological interface design (EID) framework was applied for interface design. Heuristic walkthrough by expert evaluators in a driving simulator was conducted for a proof of concept, followed by an experimental user study (n = 16) with a real-car prototype on a parking lot to demonstrate the benefits of the novel interface over an extant interface design.

Results: A novel driver interface, the virtual shadow which makes cross traffic dynamics visible through an AR HUD was proposed. A series of usability evaluation showed that the new interface outperformed the current pedestrian collision warning in driver's attention management, situation awareness, reaction with reduced workload. It also showed reduced max. deceleration and stopping distance while braking.

Conclusion: AR pedestrian collision warning that are compatible with both the driver's cognitive process and physical reality of driving, has potential benefits for driver performance and braking behavior.

Impact and Contribution: This work demonstrates the opportunity of incorporating EID into interface design for AR applications. The proposed approaches for design, prototype and usability evaluation can be leveraged by future researchers and designers to create more reliable and safer AR interfaces for vehicle drivers.

Keywords: driver vehicle interface, augmented reality, head-up display, ecological interface design, pedestrian collision warning

INTRODUCTION

According to the US Department of Transportation, in 2012 alone, 4,743 pedestrians were killed and an estimated 76,000 were injured in traffic crashes in the United States. On average, a pedestrian was killed every 2 hours and injured every 7 minutes in traffic crashes (NHTSA, 2014). Most common contributing factors for pedestrian fatality (NHTSA, 2014) suggest that drivers failed to appropriately detect pedestrians due to low visibility (16%, e.g., dark clothing, no light, etc.) or unexpected appearance of pedestrians (46%, e.g., improper crossing or being in roadway). Therefore, automakers introduced pedestrian collision warning taking advantage of recent advances in sensor technology and pedestrian detection algorithms. Once pedestrians are detected in the vehicle's path, the warning is given to the driver typically through auditory alarms or simple visual symbols. Since such warnings often lack spatial information, drivers need to further localize (i.e., recognize direction and distance of) approaching pedestrians for appropriate decision and reaction. Augmented reality (AR) heads-up display (HUD) is one of the promising technologies to address this problem by overlaying virtual graphics on top of vehicle's path to notify the driver of the specific location of detected pedestrians.

However, user interface design for AR applications has an inherently unique challenge; users must interact with not only information on the display but also environmental changes in the real world. Moreover, traditional user-centered design (UCD) approaches that focus on human-*computer* interaction, may not always adequately address the dynamic nature of human-*environment* interaction. As such, we propose to incorporate ecological interface design (EID) into AR interface design to complement UCD approaches. EID is a framework for interface design that respects dynamic, environmental constraints imposed on users' behavior. EID addresses two questions for interface design; what information should be displayed (the *content* and *structure* of information that represent physical reality of the work environment), and how that information should be presented (the perceptual *forms* of interface elements compatible with human information processing) (Burns & Hajdukiewicz, 2013).

EID has been successfully applied to many domains such as telecommunication, aviation, nuclear power plant operation, manufacturing process control, healthcare and medicine (Burns & Hajdukiewicz, 2013). In the driving domain, Seppelt and Lee (2007) designed an in-vehicle display for adaptive cruise control that presents emergent shapes depending upon the relationship between the driver's car and the lead vehicle (time to collision and timed head way). A similar approach for lane change warning revealed that EID-based designs outperformed an existing design in driver judgement accuracy and confidence (Lee, Hoffman, Stoner, Seppelt, & Brown, 2006). In spite of the documented and perceived benefits of AR, there is little research or practical efforts to incorporate EID into AR interface design. Kruit et al. adopted EID to design an AR HUD-based rally car driver support system that depicts an ideal, predicted path of the car and boundary curve to show the capability and limitation of the car. However, the effect of the new interface design on user performance was not reported (Kruit, Amelink, Mulder, & van Paassen, 2005).

We purport that EID is likely well-suited to AR interface design for vehicle drivers, since driving is a spatiotemporal task that demands drivers' appropriate information processing and responses to dynamic environmental changes. Furthermore, the EID leverages an established benefit of AR– namely, the ability to overlay information directly onto real-world objects, thereby affording direct perception of both virtual and real-world information.

Taking advantages of AR HUDs, this work applied EID framework to design a novel interface that casts virtual shadows of approaching obstacles through an AR HUD, prototyped this idea for a specific use-case of pedestrian collision warning, and evaluated usability of the new interface as compared to extant interface designs.

INTERFACE DESIGN

Identifying Information Requirements

Whereas traditional UCD approaches begin by understanding the user, EID argues that the interface design should start from examining the work environment (physical and social reality of work domain) and finish by examining users' cognitive process (mental models, strategies and preferences) especially when users' goal-directed behaviors are highly affected by dynamic environmental constraints (Vicente, 1999). Therefore, we started with a work domain analysis (WDA) to capture the physical reality of driving.

Gibson and Crooks (1938) conducted insightful and comprehensive theoretical analysis on automobile driving. They defined driving as a matter of moving toward the destination while keeping the car running within the *field of safe travel* (FoST). It is a kind of invisible tongue protruding forward along the road within which certain behavior is possible without collision. At every moment, the driver's FoST can be bounded and shaped by external or internal constraints. External constraints include stationary obstacles (e.g., road geometry), moving obstacles (e.g., other vehicles, pedestrians and animals), and legal obstacles (e.g., traffic signals, road signs and markings). The internal constraints (limitation of the car's moving capability), such as the minimum braking distance and inflexibility in sharp turns at high speed, can contract or shear off drivers' FoST. The FoST exists objectively as an actual field regardless of whether the driver perceives it correctly or not (Gibson & Crooks, 1938). It is notable that the effect of moving obstacles on the driver's FoST can be estimated by not only a projection of a moving obstacle but also the projection of the driver's own car to the point of intersection of the two paths (Gibson & Crooks, 1938).

Motivated by Gibson and Crooks's work, we analyzed the work domain of driving and represented it into a two-dimensional space that consists of an abstraction hierarchy (AH) and decomposition hierarchy (DH) as shown in Table 1. AH reveals the means-ends relationship among functional components while DH decomposes the system into physical components with part-hole relationships. In the part-hole dimension, the *system boundary* was defined as the near traffic of the car and decomposed into lower level components including the driver's own car, moving, stationary and legal obstacles. In the means-ends dimension, 'safe transportation' was selected as the reason for the system's existence *(functional purpose)*. We chose 'safe transportation' over other candidates (e.g., fast, fun, or comfortable transportation) because we are interested in collision avoidance. Safe transportation means maintaining separation from obstacles for enough FoST. For safe transportation, we should comply with the *social law* of traffic rules and *physical law* of dynamics (*abstract function*). Compliance of those laws can be accomplished by actual process of traffic control and road actors' locomotion *(generalized function)*. The system components *(physical function)* that contributed to the process are the road signs, traffic signals, roadways and road actors including the driver's own car. Finally, *physical form* of each component can be described by its perceptual appearance.

To make sure the whole system is working properly (safe transportation), we need to know the system states with measurable variables. Table 2 summarizes information requirements: contents (*which variables* should be measured) and structure (*what relationships* should be maintained for safe driving). The variables were identified by asking "How could we measure each level in AH?". At the highest level, safe transportation can be measured by enough separation or gaps (e.g., headway or time to collision) among road actors. Road actors' movement can be measured and predicted from variables such as position, velocity, and acceleration. Each road actor can be characterized by moving capability such as maximum speed, minimum braking distance, or maximum acceleration / deceleration. Finally, perceptual appearance of

Table 1. The result of work domain analysis represented in abstraction-decomposition space for safe transportation

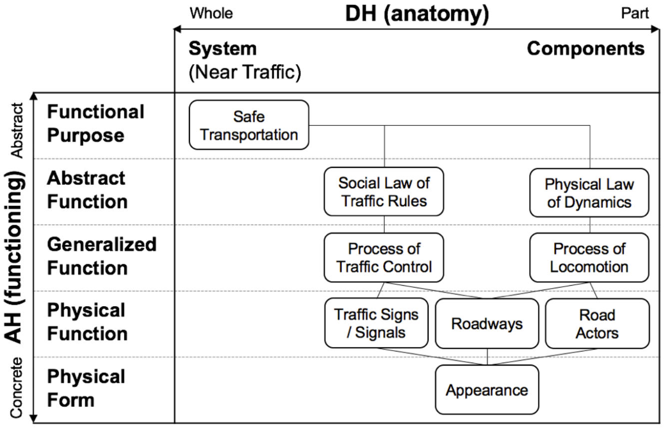


Table 2. Information requirements for safe transportation: information contents (work domain variables) and structure (relations / constraints among variables)

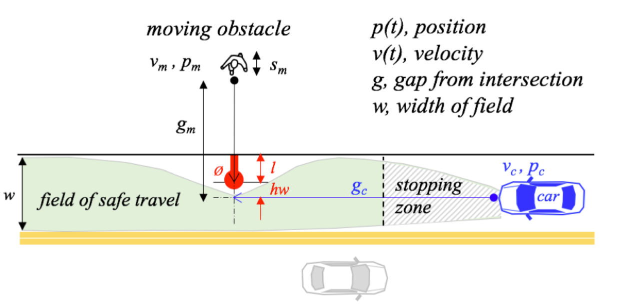
Level	Variables		Constraints: Single Variable	Constraints: Among Variables
Functional Purpose	System state variables	Spatial gap (*Headway*), *hw*	$0, HW^*, \infty$	$hw = \lvert g_m - g_c(v_m/v_c)\rvert > HW^*$ (allowable) $g_m = f(s_m, p_m, v_m)$
Abstract Function	Process governing laws			$F = ma$ $a = dv/dt = d^2p/dt^2$
Generalized Function	Process variables (components' current states)	State of traffic signals Roadway condition Position, $p(t)$ Velocities, $v(t)$ Accelerations, $a(t)$	$v < v_{max}$ $a_{min} < a < a_{max}$	
Physical Function	Components' functional capabilities	# of signal states # of lanes Linear Acc. (braking) Angular Acc. (steering)	a_{min}, a_{max}	
Physical Form	Components' appearances	Size of obstacles, s Road geometry, w	$s < s_{max}$ $w < w_{lane}$	

each component is quantified by its shape and size. Most importantly, all the defined variables cannot be out of each component's capability (*single variable constraints*) and are related to each other (*multivariate constraints*) to avoid collisions governed by physical law of dynamics.

Ecological Interface Design

Task demands of driving impose unique constraints on the driver's cognitive process; they cannot allocate all attention resources to interactions with interfaces. Therefore, we should carefully display correct information with the most appropriate ways, timing, and placement. Furthermore, outdoor use of optical see-through AR HUDs made us consider additional design factors for our interface design. All design factors can be embodied in our design metaphor as a whole.

- *Frame of reference* is one of the most important factors in AR graphics design. Graphics can be shown in exocentric (e.g., top down view) or egocentric (from the user's perspective) manner.
- *Registration (or location)* of graphics is another critical factor (Gabbard, Fitch, & Kim, 2014). Graphics can be directly attached to real world target objects (world-fixed or conformal), fixed to certain locations on the display regardless of the targets (screen-fixed) or associated with, but not directly attached to, the targets (world-associated).
- *Information density* is amount of graphics to be shown at a given time and space;
- *Shape* of graphics should embody design metaphors for ease of understanding;
- *Size* of the graphics is important for cue visibility and occlusion of drivers' field of view;
- *Color* and *brightness* can contribute to perceived meaning and visibility of graphics;
- *Intensity* (or *transparency*) of the graphics is critical especially in optical see-through AR displays not only for their visibility but also visibility of the target objects behind the graphics; and;
- *Timing* of AR interface cues is a key design factor for appropriate attention guidance and decision support.



- *headway*, $hw = | g_m - g_c (v_m / v_c) |$
- *length of tether*, l = *spatial intrusion by moving obstacle* = $w/2 - hw$
- *diameter of circle*, $ø$ = *size of moving obstacle*, s_m

Figure 1. Virtual shadow design metaphor: mapping between perceptual forms of interface elements and work domain variables & constraints. The field of safe travel and stopping zone are adapted from (Gibson & Crooks, 1938)

EID provides the SRK (Skills, Rules, Knowledge) taxonomy of human cognitive control to help designers determine how information should be displayed to be compatible with the various mechanisms that people have for processing information. A skill-based behavior (SBB) is a sensorimotor behavior based on real time processing of environmental changes with little or no conscious attention. SBB can be supported by direct perception and interaction via time-space *signals*. A rule-based behavior (RBB) is an appropriate reaction to a familiar cue in the environment based on the stored rules. RBB can be supported by one-to-one mapping between work domain constraints to *signs* in the interface. A knowledge-based behavior (KBB) requires analytic reasoning based on a mental model typically in unfamiliar situations. KBB can be supported by externalized work domain models (i.e., visualization of goal-relevant constraints) in the form of structured *symbols*. EID argues that interface design should reduce the user's cognitive load by transforming higher level cognitive demands to lower level ones while still supporting all three levels that allow users to cope with unanticipated events (Vicente, 1999).

EID helped us design a novel interface that casts virtual shadows of approaching obstacles that are immersed in the real world, taking advantage of AR HUDs. To support SBB, we present the shadow in egocentric *frame of reference* for direct perception from the driver's perspective (Figure 2). For the *registration* of AR cues, we present the shadow in a world-fixed manner such that it moves along with the target obstacle and appears larger as the driver approaches. RBB can be supported by a clear sign of collision. This is realized by associating work domain variables and constraints (identified in Table 2) with perceptual forms of interface elements (Figure 1). The *location* of the circle shows the predicted location of collision. The *shape* and *size* of a virtual shadow reflects the type and size of an approaching obstacle. The *direction* of the tether depicts the direction from which the obstacle is approaching. The *length* of the tether indicates expected spatial intrusion by a detected obstacle when the car would arrive at the intersection of the obstacle's path. The red *color* of the shadow warns the driver of an urgent situation that requires an immediate response. Finally, KBB can be supported by visualizing the dynamics of the spatial gap between the driver's car and moving obstacles

Figure 2. Prototypes of two design metaphors; bounding boxes (top) highlight detected pedestrians and the virtual shadow (bottom) visualizes dynamics of spatial intrusion by the approaching pedestrian.

over time (as the car moves). We propose that repeated use of this AR interface would help drivers develop an accurate mental model of the dynamic environment (especially with respect to drivers' time-distance judgments between own-car and moving obstacles).

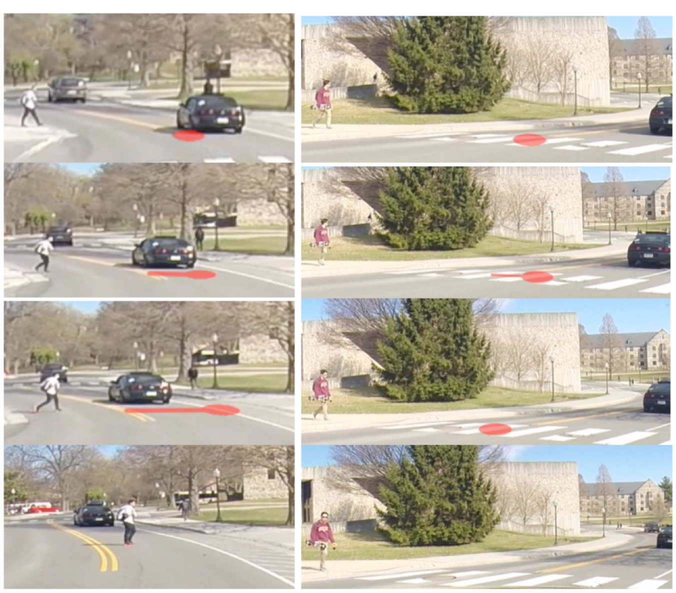
STUDY 1: ANALYTIC USABILITY EVALUATION

The purpose of the analytic usability evaluation was a validation of the new design concept with rapid prototypes before empirical evaluations with a high fidelity prototype. Therefore, we conducted a heuristic walkthrough where heuristics and representative scenarios are given to the expert evaluators (Sears, 1997).

Apparatus

We considered requirements for the fidelity of our early prototypes; The interface should interact with environmental changes while driving; The driving scenario should be representative and realistic. Users should be able to interact with the prototype in a safe driving environment. And finally, the prototype should be easy to change for design iterations.

Keeping these requirements in mind, the design idea was realized by a rapid prototyping technique using augmented video (Soro, Rakotonirainy, Schroeter, & Wollstdter, 2014). Computer generated graphics were overlaid atop pre-recorded driving video footage by a video editing tool. To explore the potential opportunity of EID-informed designs as compared to currently available driver interfaces, we prototyped this idea for a specific use-case of pedestrian collision warning. First, we prototyped an extant bounding box metaphor that highlights *any* detected pedestrians present within the pedestrian detection system's field of view (Benenson, Omran, Hosang, & Schiele, 2014). Then the virtual shadow design metaphor was prototyped with the same driving scenario (Figure 2).



(a) deceleration (b) acceleration

Figure 3. The virtual shadow in action. Two examples show how the driver's reaction affect interface dynamics (from the top to the bottom t_0->t_1->t_2->t_3). (a) the shadow starts stretching when the driver decelerates and disappears when the project of the pedestrian is beyond the vehicle's path; (b) the shadow starts shrinking back to the pedestrian and disappears when the driver accelerates so that the project of the car is beyond the pedestrian's path.

As described earlier, the spatial gap between the driver's own car and a moving obstacle is continuously changing as a function of the both moving objects' position and speed. The virtual shadow visualizes this gap by showing spatial intrusion by the obstacle (see the equation in Figure 1). Figure 3 compares two typical examples of virtual shadow dynamics. If the driver decelerates to avoid collision, the shadow of the approaching pedestrian will go further and disappear when the shadow leaves the vehicle's path (Figure 3a). On the other hand, in some cases drivers may avoid collision by accelerating the car. Then the shadow of the pedestrian shrinks back (Figure 3b) since the pedestrian will not get to the vehicle's path when the driver would pass the intersection of the two paths (pedestrian and own-car paths).

Procedure

We invited four experts (working professionals and graduate students) who meet requirements for our institution's certificate in Human-Computer Interaction and have experience in AR research. The heuristic evaluation consisted of four sessions. In the practice session, we briefly explained the procedure, interface design concepts and had experts get familiar with the driving simulator. In the walkthrough session, experts were asked to drive the car while using different interface designs (bounding box and virtual shadow). In the evaluation session, experts evaluated interfaces by predicting driver performance based on heuristics and their own expertise. Finally, in the retrospective think aloud session, we replayed driving scenarios for each interface design and let experts reflect and think more deeply about interface designs. Experts gave comments on any usability concerns, and design improvement ideas. At the end of each session, we let experts explain the rationale behind their scores by asking "which design factors (see previous section) positively and negatively affected your ratings?"

For the walk through session, augmented video combined with a high fidelity driving simulator was used (Figure 4). For a more immersive driving experience, we presented a small crosshair on the driving scene that was controlled by the steering wheel. During the driving session, we asked experts to match the crosshair to the center of their driving lane. Experts

Figure 4. A high fidelity driving simulator combined with synthesized driving video footage provided immersive driving experience and served as an appropriate tool for usability evaluation

were also asked to follow driving direction provided by voice instructions that mimic GPS navigation aids. Furthermore, we asked experts to manipulate any required controls (e.g., pedals and turn signals) as they usually drive and as prescribed by the video (e.g., when the video shows car turning, experts were expected to use the turn signals and steering wheel).

Measures

The heuristics used aimed to predict driver workload and performance at each stage of human information processing (Wickens & Hollands, 2000). Thus, experts' predictions were focused on cognitive processes such as:

- *Attention-selective*; the information would catch the user's attention quickly
- *Attention-divided*; the information would not narrow the user's attention
- *Sensation*; the information would be salient enough to be sensed against the background
- *Situation awareness-perception;* the information would guide the user's attention to the relevant elements in a given context (the visuals would help the driver detect pedestrians)
- *Situation awareness-comprehension;* the information would help the user understand the consequence of the perceived elements (the visuals would help the driver identify dangerous pedestrians)
- *Situation awareness-projection*; the information would help the user project relevant environmental elements' status into the future (the visuals would help the driver predict the dangerous pedestrians' movement)
- *Decision*; the information would help the user recognize possible reactions and the urgency of reactions
- *Workload*; the information would help reduce task demands and the user's effort to complete the task

For each of the above categories, experts were asked to give scores by predicting driver performance compared to the control condition (not having any warnings): -3 strongly worse, 0 the same, 3 strongly better than the base line.

Results

Evaluation findings suggest the virtual shadow *should* outperform the control condition in all aspects addressed by heuristics whereas the bounding box is expected to distract drivers from critical real world events and not afford reduced workload in monitoring hazardous pedestrians (Figure 5).

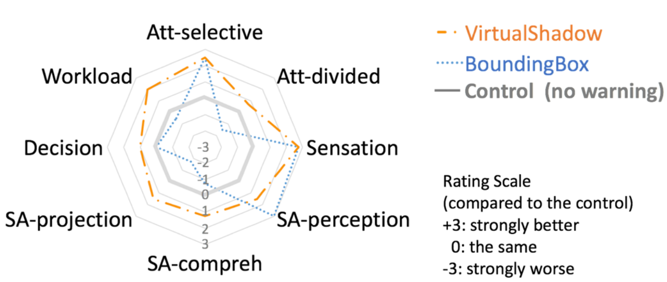


Figure 5. Predicted user performance and workload by usability experts as compared to the control condition (no visual warning)

Table 3. Mapping between predicted user performance and contributing design factors of the virtual shadow

User Perf.	Design Factors							
Process	Density	Shape	Size	Loc.	Bright	Color	Intensity	Timing
Sensation		+		+		⊕	+	+
Attention	-		⊙	⊕			+	+
Perception	+			+	+	+	+	+
Compreh.	+	⊙		⊙		-		⊕
Projection	+	-		+				+
Decision	+	-	+	+		⊙	+	+
Workload	⊕	-		+		+	-	⊕

Experts' comments during the retrospective think aloud were captured into a matrix to visualize relationships between user performance and interface design factors (Table 3). A common comment from experts was that the virtual shadow would be more comfortable and decrease mental workload, because of an effect of the appropriate timing and information density (minimal number of graphics). The positive effects of the size, position, and color were that the growth and movement of the shadow would catch the driver's attention and help with perception. Experts expected that there could be potential issues from some of the current design factors. The shape and length of a tether could be an issue if the driver cannot tell which pedestrian a tether is attached to.

Discussion

The usability evaluation revealed that the virtual shadow would likely improve driver performance at each stage of cognitive processing. Regarding driver attention, results suggest that the bounding box would guide driver attention to pedestrians but then distract drivers from other critical environmental elements by narrowing their attention. This finding resonates with the well-known tradeoff between cost (*worse divided attention*) and benefit (*better selective attention*) of attentional guidance (Wickens & Hollands, 2000) and one of the most challenging issues in AR applications (e.g., highlighting lane markers reduced pedestrian detection at nighttime driving, Sharfi and Shinar (2014). Conversely, the virtual shadow is expected to achieve these two contradicting goals by cueing only pedestrians who are expected to intrude the vehicle's path. Regarding driver situation awareness, the virtual shadow is expected to help drivers identify dangerous pedestrians and predict their movements for appropriate decision and response. With the bounding box, however, drivers would need to filter out dangerous pedestrians among the clutter and predict their movement based on drivers' experience or expertise.

More importantly, the novel design metaphor would allow drivers to accurately predict possible collisions by visualizing the invisible mechanism of collision (see the equation in Figure 1) in the form of a familiar shadow metaphor. Moreover, drivers could do so by relying on lower level perceptual process rather than high level analytic mental computation. Therefore, more attentional resources may be reserved for drivers to deploy their attention broadly across other environmental elements that might be critical or important in a given driving context. As such, the virtual shadow balances cost and benefit of attentional guidance.

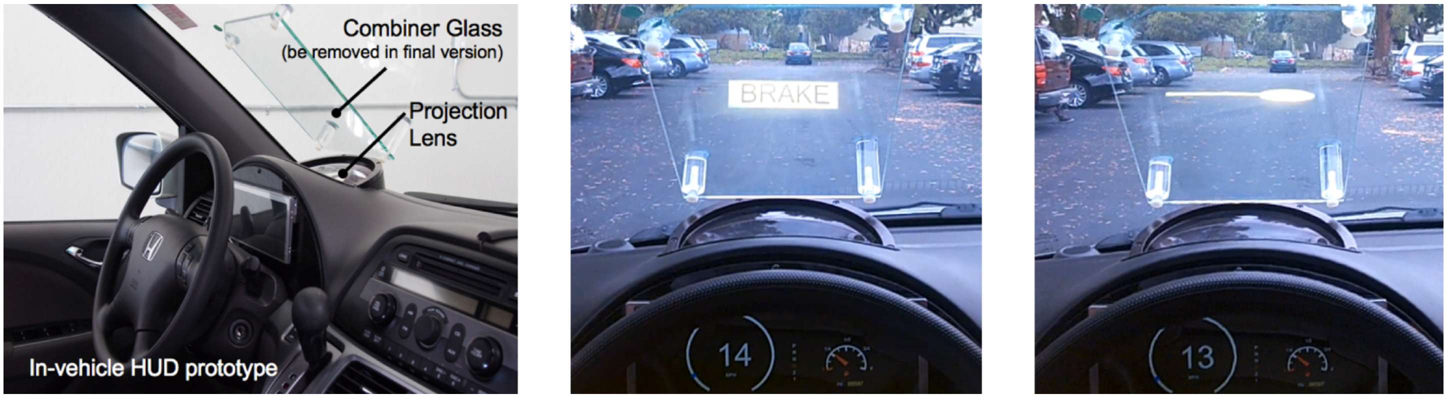


Figure 6. Two interface designs implemented in (a) the in-vehicle volumetric HUD prototype: (b) traditional warning, and (c) virtual shadow

STUDY 2: EMPIRICAL USABILITY EVALUATION

The objective of the empirical user study was to demonstrate the benefit of the new interface by answering two questions regarding pedestrian collision warning; (1) Can visual warnings on HUDs improve driver performance? (2) Can spatial information provided by AR HUDs change braking behavior? To answer those questions, we ran an experimental user study.

Apparatus

Design ideas were implemented on an in-vehicle volumetric HUD that can show computer graphics at variable depths from 8m to optical infinity with about 17° field of view (Figure 6.a). Volumetric displays form a visual representation of an object in a 3D space using multiple focal planes, as opposed to stereoscopic displays that simulate depth by presenting offset images to the left and right eye at a fixed focal plane (Blundell, 2012). Our volumetric display with time-multiplexed multi-focal planes can fast switch virtual images between far and near distance, creating a flicker-free appearance of the virtual objects rendered sequentially at different depths. This provides depth cues such as not only binocular disparity but also accurate accommodation, convergence and motion parallax without head or eye tracking. Since it projects computer graphics at the same focus distance of the target objects, drivers do not need to shift focus between virtual objects on the display and real objects in the scene. Therefore, it helps reduce perceptual (e.g., dual images and defocus blur) and safety issues caused by focal depth mismatch (Kruijff, Swan, & Feiner, 2010; Tufano, 1997). We believe 8m focus depth is reasonable for collision warning because we need to warn the driver in advance to avoid collision. For example, when we consider the generally accepted 2 second rule, the driver needs to be warned 13m ahead of the threat at 15mph. For any hazard occurring within 8m, a complementary auditory alarm could be more effective.

We used a wizard of Oz method to simulate pedestrian localization technology by experimental manipulation. We assumed our capability of detecting and localizing pedestrians; e.g., via V2P communication (Honda, 2015). To simulate this, we pre-defined locations of pedestrians and a trigger point for pedestrians to emerge. When the test vehicle passed the pre-defined trigger point, a global positioning system (GPS) triggered the visual warning to appear. This was accomplished by an excellent quality real time kinematic GPS: OxTS RT4003 with less than 20 cm localization error. This event also transmitted to the (acted) actual pedestrian by a walkie-talkie. We sent the signal to the pedestrian about a half second in advance, considering the time delay of his reaction based on our pilot test and practice.

In the traditional warning, a “BRAKE” sign is shown to notify drivers of presence of pedestrians in the vehicle’s path (Figure 6.b). It was based on head up warnings currently available in some luxury cars. Regardless of distance to the target, the sign remains the same at the center of the display until the car stops. In the new warning, the virtual shadow is shown to notify drivers of direction and distance of an approaching pedestrian (Figure 6.c).

Procedure

Sixteen experienced drivers were asked to drive a test vehicle in a parking lot and brake for crossing pedestrians while using different interface designs for pedestrian collision warning. We designed a two factor repeated measures experiment, such that participants completed the driving tasks under each of six conditions: 3 interface designs (no warning, traditional warning - brake sign, and new warning - virtual shadow), and 2 levels of distance to the target pedestrians (near vs far). We introduced two different levels of distances to the target pedestrians that might affect the driver’s perceptual risk and yield different braking behaviors. The near targets appeared when the time to collision (TTC) was 2.5 sec. The 2.5 sec TTC was set to be short enough to induce last second hard braking and selected from referencing previous braking study literature (Fitch et al., 2010). We doubled the distance for the far targets to encourage timed-normal braking. First participants were asked to drive without any warning signs for both distance conditions, which served as baseline. Then participants completed the rest of four experimental conditions. The order of interface design experienced was counterbalanced to minimize learning effects. We also randomized the location of road events and randomly added no event trials to minimize drivers’ anticipation of pedestrian presence. During each driving trials, we time stamped location, velocity, and acceleration of the test vehicle for further analysis.

In our user study, we gave participants a realistic parking lot scenario where they are asked to drive with constant speed of 15 mph until they arrived at the pre-defined parking area designated for them. While driving they were asked to brake for any crossing pedestrians. The road events prompting drivers' reaction were always actual pedestrians. In visual warning conditions, drivers additionally saw computer graphics on the HUD.

Measures

To characterize driver braking behavior and performance, we analyzed deceleration profiles during each braking maneuver and defined dependent variables (Figure 7) such as car reaction time (T_{react}), braking time (T_{brake}), time to stop (T_{stop}), maximum deceleration (G_{max}) and stopping distance (D_{stop}). We defined car reaction time as the elapsed time between the onset of a stimulus (when the car passes pre-defined trigger line) and the beginning of the car's deceleration. Braking time was defined as time spent for decelerating. Reaction and braking time add up to equal time to stop. Braking behavior was characterized by the process measures including reaction time, braking time, and maximum deceleration. Faster reaction followed by longer braking time along with smaller maximum deceleration was considered smoother timed braking behavior. Driver performance in pedestrian collision avoidance was evaluated by the final outcome of vehicle measure, stopping distance or gap from the target when the car stopped.

Results

We eliminated data from two participants who completely ignored the visual cues. Post-test interview revealed that they did not want to rely on technology, but rather believed in their own driving skill and experience. Therefore, we analyzed data from 14 participants using two-way repeated measure Analysis of Variance (ANOVA) to examine the effect of interface design and target distance. We found significant main and interaction effects on all five dependent variables, and performed post-hoc contrast tests for the planned comparisons. The results are summarized by mean and interaction plots (Figure 8).

Reaction, Braking Time and Time to Stop. Post-hoc tests (Figure 8. a and c) revealed that the virtual shadow reduced reaction time while increased braking time which resulted in the same level of time to stop compared to the baseline for both near and far targets. The traditional warning reduced both reaction time and braking time, which resulted in reduced time to stop. More interestingly, as shown in the interaction plot (Figure 8b), target distance did not affect reaction time when drivers used traditional warning.

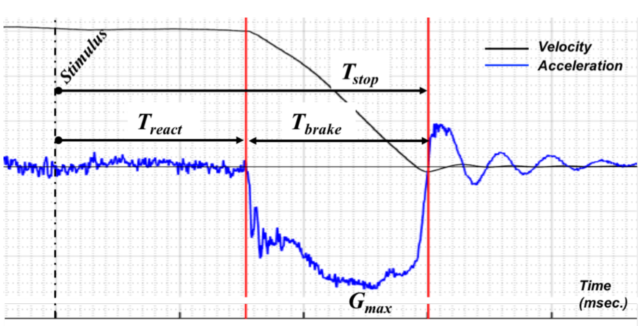


Figure 7. Dependent variables derived from deceleration profile

Maximum Deceleration. Figure 8 (e) and (f) summarizes the post-hoc test results. For the near targets, both interface design reduced maximum deceleration. For the far targets, the traditional warning showed even higher maximum deceleration than the baseline.

Stopping Distance and Gap. For the far targets, both interface designs reduced stopping distance, which resulted in larger gaps leading up to the pedestrians. The interaction plot indicates drivers had similar stopping distances (i.e., traveled distances) regardless of target distance when their using the traditional warning.

Discussion

In this study, we investigated the effect of visual warning presentation methods on drivers' performance and braking behavior in pedestrian collision avoidance. Before presenting any visual warning, we evaluated drivers' baseline performance when they did not have any warnings. We intended to run a pretest-posttest design where our main interest was behavioral changes while using warning systems compared to drivers' usual (intact) behavior. Therefore, we did post-hoc contrast tests for the planned comparisons focusing on mean differences in both performance and behavior compared to the baseline.

Regarding driver performance, both the traditional warning sign and the virtual shadow reduced stopping distances (or increased gaps from the pedestrian) for the far targets. This

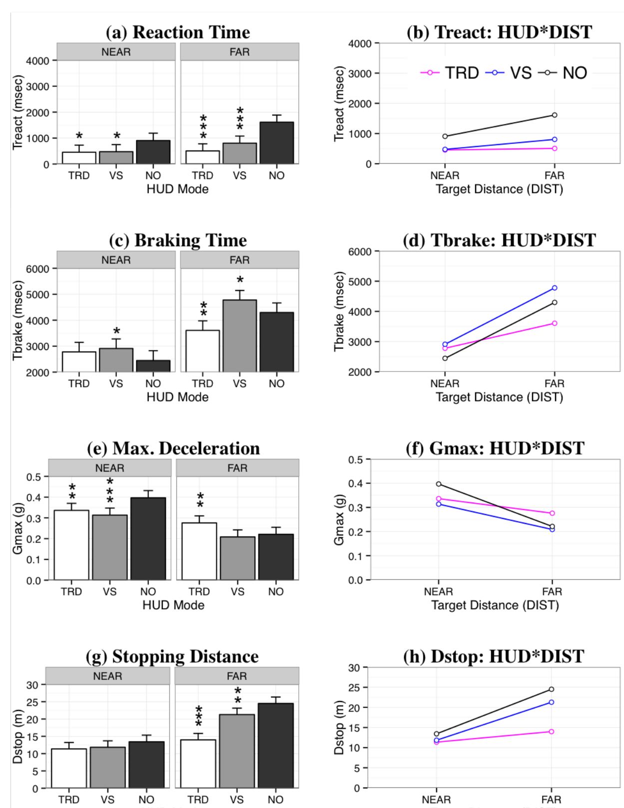


Figure 8. Driver Braking Behavior and Performance**.** Mean and interaction plots of data from 14 participants. Mean values are reported with 95% confidence intervals. Group differences are reported with *p-values* from the contrast tests compared to the baseline *('***' p < 0.001, '**' p < 0.01, '*' p < 0.05.)* TRD = Traditional warning, VS = Virtual Shadow, NO = No warning.

result suggests that visual warnings, whether they provide spatial information or not, improved driver's performance. For the near target we did not find any improvement.

Regarding braking behavior, both interface designs reduced reaction time implying that visual cues help with driver's faster detection. Interestingly traditional warnings caused even harder braking for the far target, which is not necessarily required to avoid collision. Actually drivers showed similar braking behavior regardless of target distance in response to the warning signs. This is probably because the sign lacks spatial information about approaching pedestrians. On the contrary, the virtual shadow resulted in smoother braking for both near and far targets. This result suggests that spatial information presented by conformal graphics resulted in smoother braking behavior. The trend is more obvious when we compare behavioral changes caused by target distance among three interface designs. For example, the interaction plots in Figure 8 (b) and (d) show the same differential effect of target distance between virtual shadow and no warning conditions where only stimuli for the driver were actual pedestrians. This implies that virtual shadows may provide the same level of depth cues as actual pedestrians.

CONCLUSION

This work demonstrates the opportunity of incorporating EID into interface design for automotive AR applications. We proposed a novel interface, the virtual shadow, which makes cross traffic dynamics visible through an AR HUD. The heuristic evaluation expected virtual shadow to benefit driver performance at each stage of cognitive processing: attention management, situation awareness, decision and workload. The empirical usability evaluation demonstrated that pedestrian collision warning with spatial information provided by conformal graphics on the HUD can result in not only better driver performance but also smoother braking behavior.

This work also developed realistic driving scenarios and a test protocol for usability evaluation of pedestrian collision warning systems. To the best of our knowledge, this work is one of the first usability evaluations of in-vehicle head up displays with 3D presentation capability. The proposed design and findings can be extended to other use-cases such as cross traffic alerts to avoid collision with vehicles backing up in parking lots, given that high performance object detection and localization technology is available.

ACKNOWLEDGEMENTS

This submission was based on two studies led by the author who is the only student member. Special thanks goes to Dr. Joe Gabbard (jgabbard@vt.edu) for all of his insights, time and guidance throughout much of the work; research design, analysis and writing. The first study was conducted in Apr. 2015 and will be presented at the 2016 HFES annual meeting. The author would like to thank to Jessica Isleib for her effort during the design walkthrough. The second study (conducted in May. 2015 – Aug. 2015) was supported by Honda Research Institute's internship program and has been published (Kim et al., 2016). I would like to thank Honda Motor Company for providing the test vehicle, prototyping of the proposed design idea and recruiting participants for the experiment.

REFERENCES

Benenson, R., Omran, M., Hosang, J., & Schiele, B. (2014). Ten Years of Pedestrian Detection, What Have We Learned? *arXiv preprint arXiv:1411.4304*.

Blundell, B. G. (2012). Volumetric 3D Displays *Handbook of Visual Display Technology* (pp. 1917-1931): Springer.

Burns, C. M., & Hajdukiewicz, J. (2013). *Ecological interface design*: CRC Press.

Fitch, G. M., Blanco, M., Morgan, J. F., Rice, J. C., Wharton, A., Wierwille, W. W., & Hanowski, R. J. (2010). *Human Performance Evaluation of Light Vehicle Brake Assist Systems: Final Report*. Retrieved from

Gabbard, J. L., Fitch, G. M., & Kim, H. (2014). Behind the Glass: Driver Challenges and Opportunities for AR Automotive Applications. *Proceedings of the IEEE, 102*(2), 124-136.

Gibson, J. J., & Crooks, L. E. (1938). A theoretical field-analysis of automobile-driving. *The American journal of psychology, 51*(3), 453-471.

Honda. (2015, December 15). Vehicle to pedestrian communication. Retrieved from http://world.honda.com/news/2014/4140904New-Connected-Car-Automated-Driving-Technologies-ITS/print.html

Kim, H., Miranda Anon, A., Misu, T., Li, N., Tawari, A., & Fujimura, K. (2016). *Look at Me: Augmented Reality Pedestrian Warning System Using an In-Vehicle Volumetric Head Up Display.* Paper presented at the Proceedings of the 21st International Conference on Intelligent User Interfaces.

Kruijff, E., Swan, J. E., & Feiner, S. (2010, 13-16 Oct. 2010). *Perceptual issues in augmented reality revisited.* Paper presented at the Mixed and Augmented Reality (ISMAR), 2010 9th IEEE International Symposium on.

Kruit, J. D., Amelink, M., Mulder, M., & van Paassen, M. M. (2005, 12-12 Oct. 2005). *Design of a Rally Driver Support System using Ecological Interface Design Principles.* Paper presented at the Systems, Man and Cybernetics, 2005 IEEE International Conference on.

Lee, J., Hoffman, J., Stoner, H., Seppelt, B., & Brown, M. (2006). *Application of ecological interface design to driver support systems.* Paper presented at the Proceedings of IEA 2006: 16th World Congress on Ergonomics.

NHTSA. (2014). Traffic safety facts 2012. *Traffic Safety Facts FARS/GES Annual Report*, I.

Sears, A. (1997). Heuristic walkthroughs: Finding the problems without the noise. *International Journal of Human-Computer Interaction, 9*(3), 213-234.

Seppelt, B. D., & Lee, J. D. (2007). Making adaptive cruise control (ACC) limits visible. *International Journal of Human-Computer Studies, 65*(3), 192-205.

Sharfi, T., & Shinar, D. (2014). Enhancement of road delineation can reduce safety. *Journal of Safety Research, 49*(0), 61.e61-68.

Soro, A., Rakotonirainy, A., Schroeter, R., & Wollstdter, S. (2014). *Using Augmented Video to Test In-Car User Experiences of Context Analog HUDs*. Paper presented at the Adjunct Proceedings of the 6th International Conference on Automotive User Interfaces and Interactive Vehicular Applications, Seattle, WA, USA.

Tufano, D. R. (1997). Automotive HUDs: The overlooked safety issues. *Human Factors: The Journal of the Human Factors and Ergonomics Society, 39*(2), 303-311.

Vicente, K. J. (1999). *Cognitive work analysis: Toward safe, productive, and healthy computer-based work*: CRC Press.

Wickens, C., & Hollands, J. (2000). *Engineering Psychology and Human Performance*: Pearson.